

A SYSTEMATIC REVIEW

Anticipation in Racquet Sports using Contextual Information

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ABSTRACT

The aim of the paper was to systematically review the literature on contextual sources of information for anticipation in racquet sports, examining evidence for types of contextual sources of information, methods used to evaluate it and its effect on anticipation of athletes in racquet sports. A comprehensive literature search of Scopus, SPORTdiscus, PsychINFO, PsycArticles, PubMed and Web of Science in January 2018 yielded 7 experimental studies spanning 18 years (January 2000 - January 2018). The results indicated that athletes use a variety of contextual sources of information for anticipating the outcome of an event including game score, game pattern and tactical initiative. There were also skill level differences in anticipation as skilled and older athletes were more effective in anticipating the outcome than less skilled or novice and younger players. Most of the studies used laboratory based setting using video stimuli to test these contextual sources of information. In addition, the studies found positive results of contextual information on anticipation in racquet sports. Future researchers should address the methodological limitations as well as of the sport psychology anticipation literature.

Key words

Non-kinematic, anticipatory skills, ennis, squash, contextual information

INTRODUCTION

Success in racquet sports is determined by the player's ability to anticipate the game events effectively (Rowe & McKenna, 2001). For example, if the Tennis players do not anticipate correctly, they may find themselves unable to intercept a volley or return a serve due to the time constraints of the game (Crognier & Féry, 2001). Anticipation is defined as the ability of a player to foresee the most likely outcome of an event before it occurs

(Williams, Ford, Eccles & Ward, 2011). There are two essential and related sources of information to anticipate the likely outcome of an event used by the sports person (a). kinematic or postural information of the opponent and (b). contextual sources of information (Loffing & Cañal-Bruland, 2017).

An enormous amount of research on postural source of information (i.e. bodily movements of the athlete and the racquet movements as a result of these movements)

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is well established showing the ability of skilled performers to process this information more accurately (e.g., Jones & Miles, 1978) and quickly (e.g., Williams, Ward, Knowles, & Smeeton, 2002) to anticipate an outcome of a forthcoming event than their less-skilled counterparts. Similar findings have been reported in racquet sports (see review of Mann, Williams, Ward & Janelle, 2007).

Recently, the research examining the role of non-kinematic or contextual information in anticipation has gained importance (Triolet, Benguigui, Le Runigo, & Williams, 2013; Cañal-Bruland & Mann, 2015). Context can be defined as “the set of circumstances or facts that surround a particular event or situation” (Context, n.d.). Contextual sources of information can be of different types, such as player’s position on court, knowledge of strengths and weaknesses of an opponent, different climatic conditions (Buckolz, Prapavesis, & Fairs, 1988) and knowledge of action preferences of an opponent (Mann, Schaefer & Cañal-Bruland, 2014). Several terms have been used to refer to the contextual source of information including: prior knowledge of probable upcoming events (Buckolz et al, 1988); non-kinematic (see review Loffing & Cañal-Bruland, 2017); situational probabilities (for e.g., Crognier & Féry, 2005; Ward & Williams, 2003); situational probability information or pre-event kinematic information (Abernethy et al, 2001); subjective probabilities (Bakker, Whiting & van der Brug, 1990); and contextual information (Murphy, Jackson, Cooke, Roca &

Williams, 2016).

Skilled performers are found to effectively anticipate the outcome of a forthcoming event based on contextual source of information picked up in advance of postural information becoming available (Triolet et al, 2013). They use this information to anticipate more effectively, as compared to when this information is not available (e.g., Farrow & Reid, 2012). Nowadays, performance analysts are employed by the sporting organisations to watch matches and document the probabilistic information of the opponents to support their players and coaches, with this information, in an effort to enhance their ability to anticipate (Hughes & Bartlett, 2002). Several sports have demonstrated the positive effects of providing contextual information to participants, in addition to postural information (see review, Cañal-Bruland & Mann, 2015).

However, systematic research on the contribution of contextual sources of information to anticipation in sport is limited. The evidence-based answers are required to be provided by psychological research in sports to develop theory and guide its practise. Given that these recent studies have highlighted a surprisingly large role for contextual information in supporting anticipatory behaviour, the aim of the present systematic review is to identify various types of contextual sources of information that influence anticipatory behaviour in racquet sports; because, it is essential to identify the situations and constraints under which anticipation behaviour occurs in order to develop

representative tasks that effectively capture the processes involved (Triolet et al, 2013). The second aim of this systematic review is to identify the types of methods used to study contextual information in racquet sports as well as to find out the effectiveness of this type of information on performance in racquet sports.

METHODOLOGY

The reporting of this review followed the guidelines of PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) checklist (Moher, Liberati, Tetzlaff & Altman, 2009).

Search Strategy

An electronic database search was completed in January 2018, using six relevant databases: Web of Science, SPORTDiscus, Scopus, PsycINFO, PubMed, and Psyc Articles. The search was completed for title and abstracts to identify research articles that assessed the contextual source of information on anticipation of racquet sport players; and the keywords were developed with the help of the university librarians. Keyword combinations included three groups of search terms which related to: i) the source of information (non-kinematic* OR context* OR Probabilis* OR Situation*); ii) the outcome (Anticipat*); and iii) the sport (racquet sport* OR racket sport* OR Tennis OR Badminton OR Table Tennis or Squash). Additionally, the bibliographies of relevant articles were searched and no new relevant records were identified for inclusion.

Selection Criteria

The review met the following inclusion criteria: 1) experimental design involving human beings as the participants, 2) included a component of contextual source of information being tested for anticipatory behaviour, 3) in English and published between 2000 and before January 2018 and 4) done on a racquet sport including Badminton, Tennis, Table Tennis or Squash. The studies identified via the initial search were screened for eligibility and were excluded if they did not meet the inclusion criteria. The full-text of those papers that were considered potentially relevant, following title and abstract screening, were retrieved and assessed for eligibility following full-text review. A PRISMA flow diagram of the selection process is provided (Figure 1).

Performance outcomes included objective measures. The studies which examined only the effect of kinematic sources of information on anticipatory skills as well as studies which examined the effect of non-kinematic sources of information on anticipation apart from racquet sports were excluded. Studies not measuring component of non-kinematic sources of information were also excluded.

Data Synthesis

The following data about the author, participant characteristics, sport, location, methods used, purpose of the study, performance measures used and the findings was extracted and collated from each of the included articles (Table 1).

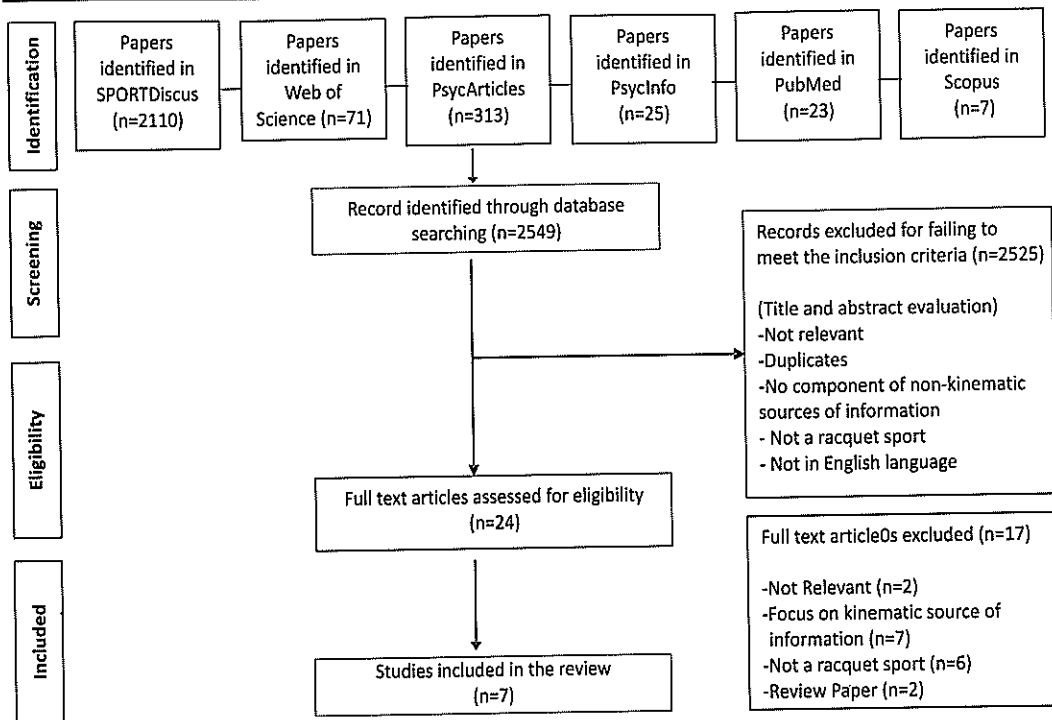


Fig.-1: PRISMA flow diagram outlining the implementation of the systematic search strategy and review process.

Table-1: Showing collected data from included articles

Author	Farrow & Reid (2012)
Sport	Tennis
Participants	15 Males (Older Group); Mage=17.93 (S.D.=3.30); Australian Tennis rank between 2 and 241. 14 Males (Younger Group); Mage=11.29 (S.D.=0.45); Australian Tennis rank between 329 and 1606.
Location	Australia
Aim	The purpose of this study was to determine the contribution of event probability information to the anticipatory responses of skilled Tennis players representative of two different ages and stages of development.
Contextual Information	Relationship between Game Score (First point of each game) and resultant serve location.
Method	Laboratory based (Using Plasma Screen Touch)
Performance Measure	Response Time and Response Accuracy, Awareness of existence of advance probabilistic information.

Findings	<u>Response Time</u> Older players picked up the occurrence of contextual information after ninth service game, whereas younger players did not. A significant age group by game interaction for response time ($F(11,297)=3.86, p<.05, \eta^2=.12$). Older players initiated their response 900ms before racquet-ball contact from 9th to 11th service game and 1300ms before racquet-ball contact in the 12th service game. <u>Response Accuracy</u> Response Accuracy did not improve for any of the two groups as there was no significant age group by game interaction effect ($F(11,297)=.96, p>.05, \eta^2=.01$). <u>Awareness of Contextual Cues</u> 14 out of 15 older players were aware of repeating the first serve location on the first point, whereas only 1 out of 14 younger players were explicitly aware of this information.
Author	Abernethy, Gill, Parks & Packer (2001)
Sport	Squash
Participants	6 Males (Experts); Age 18 to 27 years (Mage=20.5 years); Australian institute of Sport Squash unit or top players within the highest level of district competition with a playing experience of 5 to 15 years. 6 Males (Less-Skilled); Age 18 to 26 years (Mage=21.3 years); Regular Squash Players in the lowest level (division D3) of local Squash competition with a playing experience of 1-10 years.
Location	Australia
Aim	To examine the contribution that attunement to the information contained in global event probabilities and sequential dependences within the opponents' pattern of play.
Contextual Information	Game pattern
Method	In-situ (Using PLATO liquid-crystal occlusion glasses)
Performance Measure	Direction and depth of opponent's stroke.
Findings	Expert players performed better in the condition of maximum occlusion (620ms before the racquet-ball contact) with respect to less skilled players due to the perception of the information contained in the situational probabilities of the game pattern of their opponents. Expert players showed less amount of direction and depth errors with respect to opponent's stroke in all occlusion conditions (The prediction error of the expert group was around 27% (average) and 48% (average) for the less-skilled group from 220ms prior to racquet-ball contact to as early as 620ms prior to this contact suggesting an increase in Response Accuracy for both groups).

Author	Löffing & Hagemann (2014)
Sport	Tennis
Participants	26 males (skilled); Mage=24.69 years (S.D.=4.29); Tennis playing experience: M=16.65 years, S.D.=4.72; individual's highest performance level: second to seventh highest league in Germany.
Location	Germany.
Aim	To examine the effect of the opponent's shot position on anticipatory responses of Tennis players.
Contextual Information	On-court position
Method	Laboratory based experiment (Using Point Light Delays)
Performance Measure	Response Accuracy and Response Time
Findings	Shot position by temporal occlusion interaction for skilled players was significant ($F(6,150)=9.17$, $p<.001$, $\eta^2=.27$) and non-significant for novice players ($p=.61$). RT decreased for expert tennis players due to situational probability information of the on-court position of the opponent. Expert Tennis players had a higher Response Accuracy and lower Response Time than novice players.

Author	Crognier & Féry (2005)
Sport	Tennis
Participants	17 Males (experience players); Mage=23 years (S.D.=2.6); Tennis playing experience: M=15 years; Competitive matches played: M=465 (S.D.=133)
Location	All participants has normal (uncorrected vision)
Aim	France To examine the ability of tennis players to anticipate their opponent's stroked when the player has the opportunity to impose his or her playing intent
Contextual Information	Tactical initiative (reflecting the possibilities of controlling rallies)
Method	In-situ (using PLATO liquid-crystal occlusion glasses)
Performance Measure	Directional Response Accuracy of prediction of the site at which the ball arrived.
Findings	Directional estimation was more accurate in High Initiative Situation (HIS) conditions than in Moderate (MIS) or Weak Initiate Situation (WIS) conditions ($T=3.32$, $p<.05$ and $T=3.41$, $p<.05$) respectively. A significant difference for Directional Response was found only for HIS ($T=3.62$, $p<.05$). Participants were best able to identify the correct direction in which to move to intercept the ball in HIS conditions.

Responses on horizontal precision were more accurate in HIS than in MIS conditions; $F(1,16)=15.48$, $p<.05$ or WIS $F(1,16)=12.30$, $p<.05$.

Author	Murphy, Jackson, Cooke, Roca, Benguigui & Williams (2016)
Sport	Tennis
Participants	16 Males (Skilled Tennis players); Mage=24 years (S.D.=5.6); Tennis playing experience: $M=17.8$ (S.D.=5.5); Number of hours of Tennis in their career per week: $M=13.3$ (S.D.=3.6); Competitions per year: $M=19.1$ (S.D.=7.3) 20 Males (Less-skilled tennis players); Mage=24.1 years (S.D.=4.7); Tennis playing experience: $M=7$ (S.D.=4.8); Number of hours of Tennis in their career per week: $M=1.4$ (S.D.=0.6); Not competed in competitive tennis tournaments outside of internal staff box leagues of school Tennis.
Location	UK
Aim	To investigate the extent to which skilled and less-skilled Tennis players use contextual information to anticipate.
Contextual Information	Sequential relative movements of the opponent and ball flight information
Method	Laboratory based (from a third person perspective
Performance Measure	Response Accuracy
Findings	Response Accuracy of both groups (skilled and less skilled) were significantly higher than would be expected due to chance in both normal video and animated display conditions when making depth, direction, and combined anticipation judgments (all $p<.001$).
Author	Broadbent, Ford, O'Hara, Williams & Causer (2017)
Sport	Tennis
Participants	21 (intermediate level) who were divided into 2 groups: sequential practice ($n=11$; Mage=20.7 years, S.D.=1.6) or a non-sequential practice group ($n=10$, Mage=20.9 years, S.D.=1.1)
Location	UK
Aim	To examine the retention and transfer benefits of a sequential, compared to a random, structure of opponent shots during video-based anticipation training in Tennis
Contextual Information	Sequential Responses
Method	Laboratory based
Performance Measure	Response Accuracy and Response Time
Findings	The sequential group significantly reduced their response time from pre-test to transfer test. Response Accuracy was higher in sequential group as compared

to the non-sequential group from pre-test to transfer test, which indicated that contextual information on sequential practice enhanced the response accuracy of intermediate Tennis players

Author	Loffing, Solter, Hagemann & Strauss (2016)
Sport	Tennis
Participants	20 Males (skilled); Mage=30.45 years, S.D.=4.65; Tennis playing experience; M=20.05 years, S.D.=7.08 20 Males (Novice); Mage= 24.60 years. S.D.=1.96; No experience in competitive tennis All participants were right handed and reported normal or corrected-to-normal vision.
Location	UK
Aim	To examine the effect of opponent's on-court position and handedness on anticipatory skills of Tennis players.
Contextual Information	On-court position
Method	Laboratory based
Performance Measure	Prediction Accuracy and Post experiment questionnaire to measure the strategies used to anticipate the outcome.
Findings	Skilled Tennis players were affected by an opponent's on-court position across temporal occlusion conditions compared to novice players. Post experiment questionnaire revealed that nine out of sixteen skilled players used on-court position as a relevant cue to predict stroke direction, while only 1 out of 10 novices indicated the use of this cue for informing anticipatory response.

RESULTS & DISCUSSION

A total of 2549 research articles were screened through database searching. Out of these, 24 articles were screened and assessed for full text after removing duplicates and article titles and abstracts not adhering to the inclusion criteria. A further review then excluded 17 articles for the following reasons: not relevant ($n = 2$), focus on postural source of information ($n=7$), employing contextual source of information to understand anticipatory behaviour of athletes ($n =6$) and review papers ($n=2$) (See Figure 1 for a breakdown of excluded articles).

The seven included articles studied

racquet sports including Tennis and Squash. The studies included all male participants who were right handed, had a normal or corrected vision and were located in Australia, United Kingdom, Germany or France. Objective performance measures including Response Accuracy and Response Time were exclusively used to measure anticipation in all studies, except two studies (Murphy et al, 2016; Loffing, Solter, Hagemann & Strauss, 2016) measuring only response accuracy as a measure of anticipatory behaviour. The awareness of situational probabilistic information was measured through post-experiment questionnaires in two studies (Farrow &

Reid, 2012; Loffing et al, 2016) to get a qualitative support of the results achieved through the quantitative experiment. Given the various sport levels of participants, methods and type of contextual information used, the results for anticipation skills could not be aggregated. These results are presented in a descriptive manner below.

Types of Methodology

Experiments were conducted by all the seven studies to assess the effect of contextual information on anticipatory behaviour. Five studies were conducted in the laboratory-based setup and using videos as the test stimuli for the experiment. Two studies were conducted in the natural setting, i.e. in-situ (Abernethy et al, 2001 and Crognier & Féry, 2005).

Types of Contextual information in Racquet sports

Only game pattern was assessed as a source of contextual information in the sport of Squash (Abernethy et al, 2001). While variety of contextual sources of information were assessed in Tennis including relationship of game score with serve location (Farrow & Reid, 2012), on-court position (Loffing & Hagemann, 2014; Loffing 2016), tactical initiative (Crognier), sequential practice (Broadbent) and sequential relative movement of the opponent and ball flight information (Murphy et al, 2016).

Response Accuracy

Analysis of all the studies ($n=7$), revealed an inclusion of Response Accuracy as a measure of anticipation. Five studies depicted a positive effect of Response Accuracy on anticipation due to contextual information (Abernethy, Gill, Parks & Packer (2001), Loffing & Hagemann

(2014), Crognier & Féry (2005), Murphy, Jackson, Cooke, Roca, Benguigui & Williams (2016), Broadbent, Ford, O'Hara, Williams & Causer (2017), Loffing et al, 2016) and one study (Farrow & Reid, 2012) did not obtain a significant effect of situational probability of game score on anticipatory skills for any of the groups (older or younger) due to a 'ceiling effect'.

Response Time

Response time was measured as a performance measure for anticipation in five studies (Farrow & Reid, 2012; Abernethy et al, 2001; Loffing & Hagemann, 2014; Crognier & Féry, 2005; Broadbent et al, 2017) indicating a decrease in response time to anticipate the outcome on the basis of the contextual information provided in the experiment. While, Murphy et. al (2016) and Loffing et al (2016) did not measure Response time as a measure of performance outcome for anticipation.

Awareness of contextual information

Two studies used post experiment questionnaires to assess the awareness of the present contextual sources of information to anticipate to the outcome of the event by asking them to write down the strategies they used to anticipate during the experiment. This information was then quantified by calculating frequencies and reported that the contextual information was used as a source of information by majority of skilled sportsperson as they were aware of this information (Farrow & Reid, 2012; Loffing et al, 2016).

This systematic review was conducted to explore the methods and types of contextual sources of information used to anticipate the outcome of an event in racquet sports as well as to examine the effect of

these sources of information on improving anticipatory skills. Five out of the seven studies reported significant increase in Response Accuracy, with the others suggesting a non-significant effect on this component of performance measure. All studies reported a significant decrease in Response Time, and demonstrated an effect of contextual information in anticipating an outcome of an event. However, five out of the seven studies used two comparison groups including expert vs less-skilled or novices and older vs younger as well as adopted a video-based approach using laboratory setting as their methodology, whilst the other two studies used one group for the experiment including an intermediate or experienced group of players and adopted an experimental approach in the natural setting.

None of the studies had a control group in their study. Therefore, a conclusive evidence over the effect of contextual sources of information in racquet sports based on this review cannot be produced. However, a recommendation over the future research examining these effects on anticipation of an outcome event in light of several positive results can be made. Specifically, experimental research that employs a control group, greater rigour and field-based setting to determine transferability of experimental learning on the field, causality or clear intervention effect in the experiment.

The study quantified the influence of contextual information on the motor behaviour of athletes according to their sporting level. The results confirm that the manipulation of contextual information has an influence on sports performance

according to the sport level of the participants. This, in turn confirm that expert athletes, in uncertain and temporally constrained sport situations, make use of sources of early information to manifest a more anticipatory and precise behaviour (i.e., they initiate their movement before the information about the movement appears, see Raab, 2012). In contrast, players with less experience at sports level could show a more reactive than anticipatory behaviour due to a lack of knowledge of the advanced visual indexes of the offensive game sequence or the negative cost of a wrong anticipation for the team (Granda, 2002). Farrow & Reid (2012) made a comparison based on the stages of development, comparing younger players with older ones, resulting in older players taking less time to anticipate the outcome event using contextual information given to them by the game score. Since, the temporal demands of the game do not reach a stage where anticipation is required to succeed for novice or younger players (McPherson, 1999), it might not be useful to compare younger, novice or less-skilled players' ability to anticipate with the older or experienced players. Younger or novice players' lack of experience might also result in a bias when comparing this capacity with expert players (Weissensetiner, Abernethy & Farrow, 2008). Two studies were done only on single skilled level participant group without the control group, restricting the generalizability of results.

Moreover, the absence of control groups, in all the seven studies, eliminated the ability to analyse what sort of impact did contextual information had on anticipation. Using male participants solely to study the

effect of contextual sources of information in racquet sports restricted the generalizability of results based on gender. Also, the locations where the research has been conducted was limited to 4 countries; UK, Australia, Germany or France. The future researchers need to consider the diversity based on culture and gender.

Five of the studies used a laboratory based occlusion studies which was solely based on videos. It was seen that video provided a less than perfect simulation of the perceptual information of the natural world. In using display simulations, degradation of a number of (possibly critical) elements of the perceptual array is inevitable. There is loss of stereoscopic depth information (as a three-dimensional display is projected to a two-dimensional one), reduction in visual field and, frequently also, object/image size, and loss of some display resolution. It is extremely difficult in the laboratory setting to reproduce the same time stresses that exist in responding in the natural task as it removes the requirement for the movement response to be organised in time to permit coincidence/interception with the approaching ball. Typical laboratory tasks present each stimulus pattern-response option with equal probability; thus, removing event probability as a potentially potent source of anticipatory information. Given these differences between the laboratory occlusion tasks and the natural task, considerable caution is needed in assuming, generally, that the findings with respect to expertise in the laboratory context necessarily hold true in the natural setting (Abernethy et al, 2001). While, only two studies were devoid of these limitations due

to the in-situ setting of their experiment. Thus, future researchers should be cautioned in using video based temporal occlusion experiments eliminating the transferability of skills to the natural field.

The studies in this systematic review shows a limited amount of research done on anticipation in racquet sports using contextual sources of information; and it is well known that contextual sources of information play a significant role in anticipation in sports (e.g., Abernethy et al, 2001; Raab, 2012; de Oliveira et al, 2014). Majority of the studies are conducted in Tennis and only one on Squash, indicating a lack of research evidence in other racquet sports. Also, the variety and numerous studies examined the effect of contextual information on anticipating the outcome of the event in various different types of sports including contextual sources like knowledge of action preferences (Mann, Schaeffers & Cañal-Bruland, 2014; Navia, Van der Kamp & Ruiz, 2013), familiarity with a particular opponent (McRobert, Ward, Eccles & Williams, 2011), familiarity with a particular action (Milazzo, Farrow, Ruffault & Fournier, 2016) and knowledge of general action occurrence (Cañal-Bruland, Filius & Oudejans, 2015; Muller, Abernethy & Farrow, 2006). A few research studies have been conducted to directly investigate the usage of contextual information in sports domain, particularly in racquet sports, and there is a lack of empirical evidence to support the idea that they can be used to improve performance (Muller & Abernethy, 2012). Therefore, future researchers should consider these types of contextual information used in various different sports and apply it in

racquet sports to increase the evidence of various types of contextual information used for anticipation and its effectiveness.

However, the manipulation of contextual information in the experiments are based on repetition and they may result in a negative effect on anticipation and further affect the sports performance, if contextual information does not correspond to the expected outcome of the action i.e. it will be costly to the player (see Cañal-Bruland & Mann, 2015). Hence, future researchers should consider the types of contextual information studied in racquet sport to improve anticipatory performance, and should use in-situ setting for experiment conduction, enhancing the transferability of information to the field.

It was evident from the sport psychology anticipation literature that contextual source of information plays a significant role in the anticipatory behaviour of skilled players in racquet sports (e.g., Abernethy et al, 2001). It is evident that time constraints demand anticipatory skills to be successful in sports at an expert level (Triolet et al, 2013). Hence, indicating that future researchers should address the significance of this type of behaviour in different groups based on large samples, broadening the view of anticipation in racquet sports (see review Cañal-Bruland & Mann, 2015).

This review followed PRISMA guidelines in an attempt to be objective and systematic in its approach (Moher et al, 2009). The limited number of studies and variety of contextual information examined in racquet sports limits generalization. Studies were diverse in their sporting levels

of athletes and types of contextual information provided, thus making comparisons difficult. Furthermore, sample sizes were small and consisted of only male participants, therefore limiting reliability. Additionally, as research into anticipatory behaviour due to contextual information is growing, a further review comparing these findings with other sports is suggested. This review will provide novel insights for the coaches, performance analysts, and players to help them to enhance performance (Loffing, Hagemann & Farrow, 2017).

CONCLUSION

The aim of this paper was to describe the recent work spanning in the last 18 years (2000-2018) on contextual information in enhancing anticipatory skills of players in racquet sports and thereby inspiring promising future routes for research on anticipation in racquet sports and its application in practice. It should be noted that only seven studies have been used to quantify the effect of anticipatory skills using manipulation of contextual information in racquet sports. The studies in this review do not provide a conclusive evidence, primarily due to variety of methods and types of contextual information examined in racquet sports, but the evidence did show an increase in performance outcomes in racquet sport players.

However, more research is recommended to get a broader view of anticipation in racquet sport; specifically, further experimental research with control groups and methodological rigour, in order to establish correlations or significant effects.

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